

Physics-Guided Counterfactual Fault Synthesis in Cyber-Physical Systems

*Milestone 1 Submission

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Abstract—The deployment of autonomous Cyber-Physical Systems (CPS) is critically bottlenecked by the inability to rigorously validate safety against rare, catastrophic edge cases. While Generative AI offers a pathway to synthesize this missing failure data, standard models frequently hallucinate, generating scenarios that violate fundamental thermodynamic and kinematic laws. To address this, we propose a Physics-Guided Generative World Model for Counterfactual Fault Synthesis. Our methodology explicitly disentangles deterministic low-frequency macrodynamics (environmental context and underlying kinematics) from high-frequency sensor noise. We achieve this through a modular architecture integrating a Time-Series Joint-Embedding Predictive Architecture (TS-JEPA) for macro-state encoding, a deterministic decoder for baseline reconstruction, and a Conditional Variational Autoencoder (CVAE) for high-frequency jitter synthesis. Crucially, counterfactual generation is driven by a Stochastic Differential Editing (SDEdit) loop, guided iteratively by physical constraint residuals from a Physics-Informed Neural Network (PINN). We validate our approach using industrial rotating machinery (MaFaulDa dataset) as a representative CPS case study. This framework provides a scalable Generative Digital Twin, ensuring synthetic edge-case data is both statistically realistic and physically trustworthy.

Index Terms—Cyber-Physical Systems, Physics-Informed Neural Networks, Generative AI, Digital Twins, SDEdit

I. INTRODUCTION AND LITERATURE REVIEW

The health monitoring and safety validation of safety-critical Cyber-Physical Systems (CPS), such as autonomous vehicles and industrial robotics, rely heavily on the availability of rich, diverse fault data. However, collecting real-world catastrophic failure data is both dangerous and expensive. While data-driven generative models theoretically offer a way to synthesize these missing edge cases, they frequently struggle to maintain physical compliance in industrial settings. Current models are prone to hallucinating physical impossibilities, such as violating friction limits or energy conservation, rendering the generated data useless for rigorous predictive maintenance and safety validation.

Our foundation relies on recent paradigm shifts in Scientific Machine Learning (SciML) to bridge the gap between deterministic physics engines and noisy edge sensor data. We build upon our previous work [1], which established a robust fault diagnostic pipeline using a Physics-Informed Neural Network

(PINN) to generate highly informative residuals, analyzed by Extreme Value Theory and a Siamese Neural Network (SNN). To pivot from diagnosis to *synthesis*, we incorporate and disentangle Time-Series Joint-Embedding Predictive Architectures (TS-JEPA) [2], which excel at semantic representation learning by filtering unpredictable high-frequency variations, and Conditional Variational Autoencoders (CVAEs) to explicitly model high-frequency jitter.

Recent literature has increasingly explored physics-guided generative models for complex systems. *FaultDiffusion* [3] demonstrated the efficacy of diffusion models for few-shot fault time-series generation, though it relies on static class-label conditioning rather than strict physical constraints. Works such as *PhysDiff* [4] and *PINFDiT* [5] have pioneered physics-guided diffusion by projecting denoised steps into physical manifolds via Langevin dynamics. Similarly, *CoNFILd* [6] utilized latent diffusion models for spatiotemporal data, though its guidance remains largely empirical. Furthermore, Zhang et al. [7] validated the necessity of frequency decomposition in time-series diffusion, synthesizing low-frequency patterns and high-frequency details in distinct stages.

The novelty of our proposed framework lies in its architectural synthesis of a Generative Digital Twin. Unlike existing methods, we employ a Stochastic Differential Editing (SDEdit) [8] loop operating directly within a disentangled latent space. Using the PINN’s residual (that highlights violations from healthy operation), and the comprehensive embedding space of fault fingerprints in [1] as a trustworthy, continuous, differentiable penalty signal at each denoising step, we transform generation into a physics-aware steering force, strictly preserving the original operational context while synthesizing physical faults.

II. PROPOSED METHODOLOGY

The proposed methodology is a hybrid, multi-stage architecture designed to ingest a normal CPS sensor trace and generate an approximated, physically realistic faulty trace of a specified target class. The pipeline, illustrated in Fig. 1, operates in two primary phases:

Phase 1: Modular Training and Component Freezing.

This contrastive fine-tuning stage establishes the generative foundation. A TS-JEPA encoder is trained to map raw sensor traces into a low-dimensional latent space, capturing the deterministic macro-envelope (kinematics and context) while discarding high-frequency sensor jitter. A deterministic Decoder (Decoder 1) learns to reconstruct the clean envelope, while a CVAE (Decoder 2) learns to synthesize the chaotic, high-frequency fault transients (jitter) conditioned on the residual errors.

Phase 2: Deployment and SDEdit Transformation.

At runtime, a healthy trace is encoded into a latent state. Using an iterative SDEdit loop, forward noise is injected and subsequently denoised. Crucially, at each denoising step, the temporary latent state is reconstructed and passed through a frozen, pre-trained Stage 1 PINN and the feature extraction in Stage 2 [1]. This generates a physically-informed embedding space of the signatures of behavioral traces, naturally clustering faults. The log-likelihood distance of the temporary latent state in this embedding space when compared with the probability distribution of the target fault acts as a penalty, generating gradients that backpropagate to safely “bend” the healthy context into a faulty one without destroying the operational baseline. The final synthetic sensor stream is synthesized via superposition of the deterministic envelope and the generated jitter.

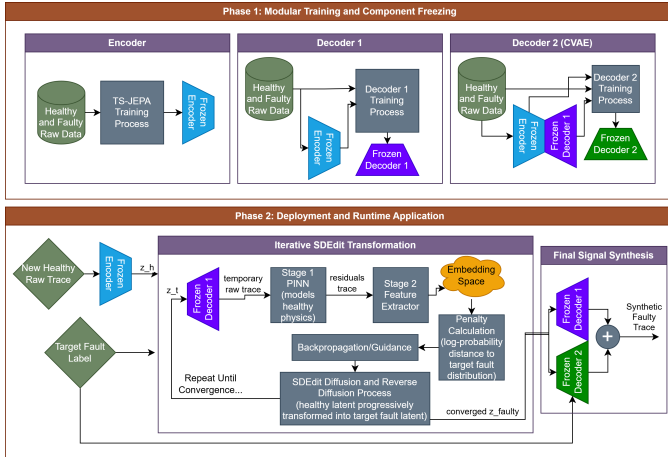


Fig. 1. Proposed Methodology: Physics-Guided Counterfactual Fault Synthesis, detailing the offline component training (Phase 1) and the iterative SDEdit latent diffusion loop (Phase 2).

III. CASE STUDY AND EXPERIMENT PLAN

A. Case Study: Rotating Machinery Dataset

To validate the generalized CPS framework, we will utilize industrial rotating machinery as a highly dynamic, physically constrained subsystem case study. We use the MaFaulDa benchmark dataset, a rich collection of high-resolution (50 kHz) vibration signals captured from a SpectraQuest Machinery Fault Simulator. Containing nine distinct fault types across varying rotational speeds (12 Hz to 61 Hz), it is

ideal for testing operational context preservation and kinematic compliance.

B. Experiment Plan

We will leverage the validated codebase from our prior study [1], re-using the pre-trained Stage 1 PINN (for healthy physical dynamics modeling), and the Stage 2 feature extraction, actively for the proposed method, and finally the Stage 3 SNN for validation of the generated synthetic data. The implementation will be executed in a phased approach:

- **Step 1: Offline Component Pre-training (Phase 1).** We will sequentially implement and train the TS-JEPA encoder, Decoder 1, and the CVAE. We will evaluate this phase by assessing the TS-JEPA’s ability to successfully disentangle low-frequency kinematics from high-frequency environmental/sensor noise, and assessing the reconstruction quality and training stability of the decoders.
- **Step 2: Integration and SDEdit Loop (Phase 2).** Once the modular blocks are frozen, we will implement the SDEdit latent diffusion process. We will systematically evaluate the computational feasibility of the Vector-Jacobian Product (VJP) gradient calculations required for the physically-guided penalty loop.
- **Step 3: End-to-End Evaluation.** The synthesized counterfactual traces will be evaluated for physical fidelity (verifying the reduction of physical hallucinations) and passed through our pre-existing Siamese Neural Network Stage 3 to measure diagnostic utility (if the synthetic faults are classified correctly by the diagnostic pipeline, and if a diagnostic pipeline trained only on synthetic data can diagnose real faults).

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